

JKBOSE 2026



CLASS 11TH

Cell Cycle & Cell Division

LECTURE-3

ZOOLOGY

DAVOOD SIR (CEO of NG)

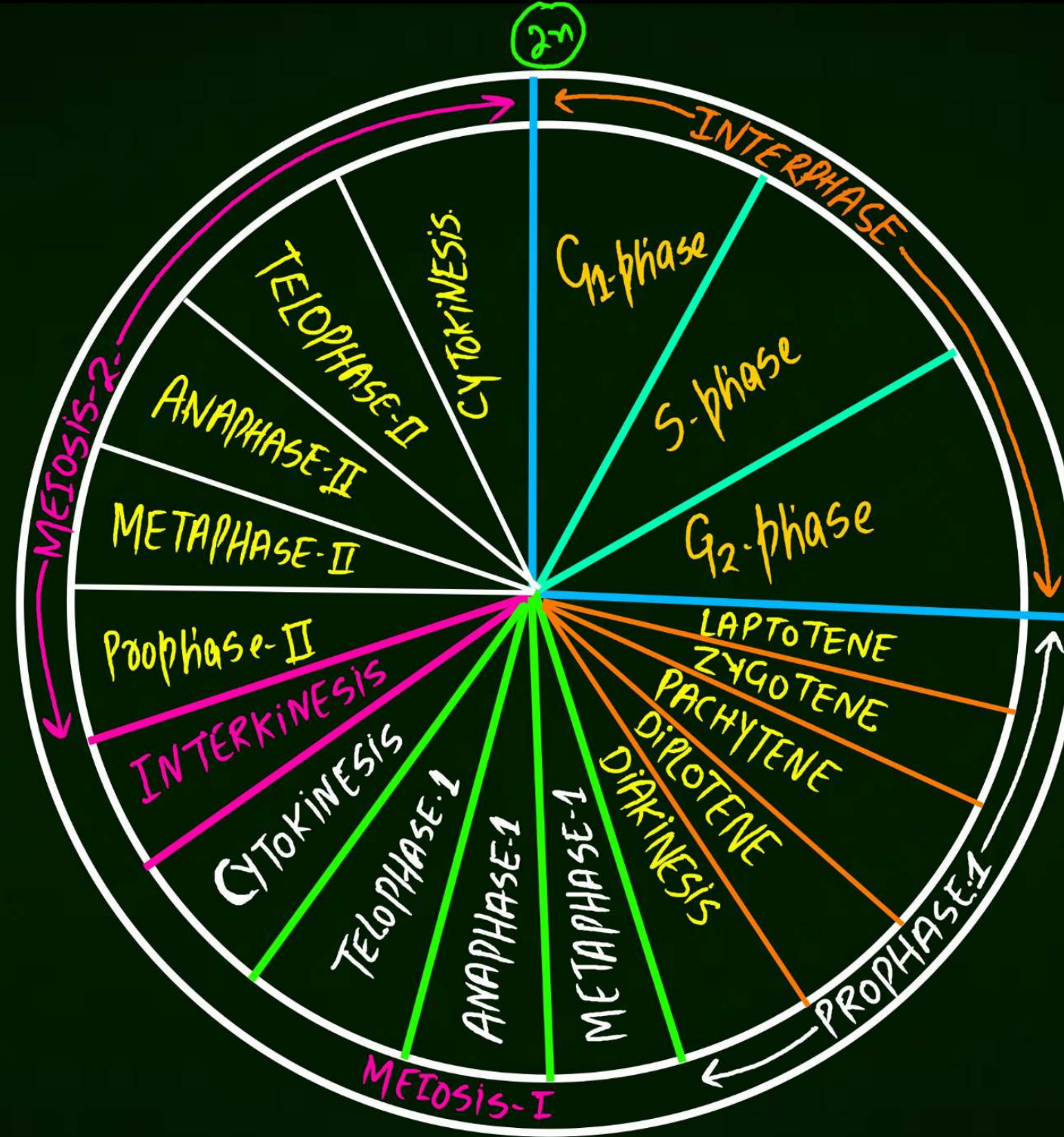


Physics Wallah

Topics To Be Covered

- 1 Meiosis (Meiosis II) ✓
- 2 Significance of Meiosis
- 3 Previous Year NEET MCQ'S

Meiosis



Meiosis II

- Meiosis II is initiated immediately after cytokinesis, usually before the chromosomes have fully elongated.
- In contrast to meiosis I, **meiosis II resembles a normal mitosis (Meiotic mitosis)**.
- Meiosis II is Equational division.

Meiosis II

PROPHASE-II

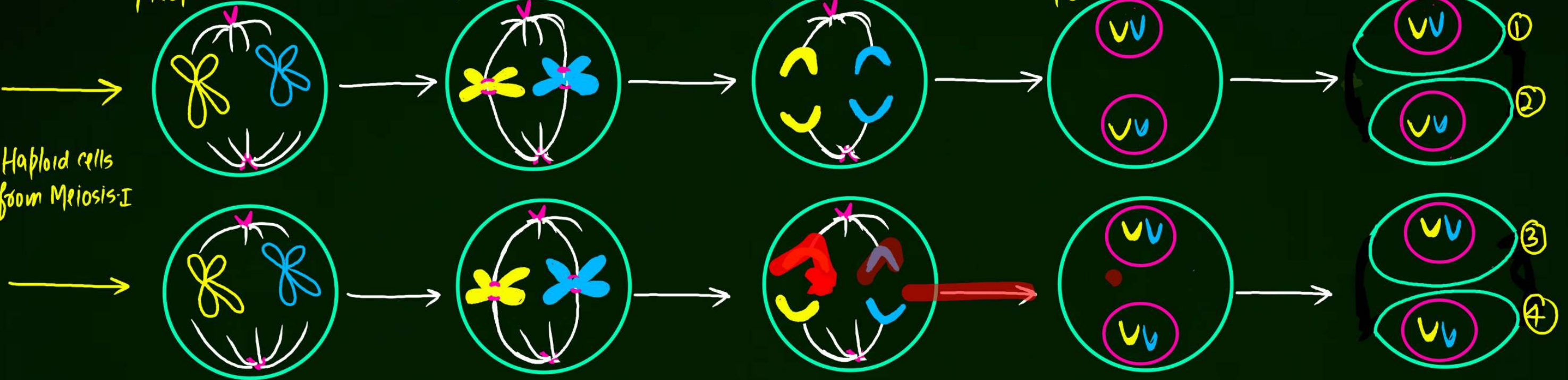
METAPHASE-II

ANAPHASE-II

TELOPHASE-II

CYTOKINESIS

Haploid cells
from Meiosis-I



- Chromosomes again become compact
- Centrosomes move to opposite poles
- Spindle fibre formation

- Chromosomes align at equator.
- Spindle fibre get attached to the kinetochore of sister chromatids.

- Splitting of centromere of each chromosome.
- Sister chromatids move to opposite poles.

- Nuclear membrane is formed which encloses two chromatid groups and results in the formation of two haploid nuclei

- Results in Four haploid cells (Tetrad)

Meiosis II

- 1) **PROPHASE II:-** The nuclear membrane disappears by the end of prophase II. The chromosomes again become compact. Centrosome move to opposite poles and spindle formation takes place.
- 2) **METAPHASE II:-** The chromosomes align at the equator and the microtubules from opposite poles of the spindle get attached to the kinetochores of sister chromatids.
- 3) **ANAPHASE II:-** Splitting of the centromere of each chromosome (which was holding the sister chromatids together), allowing them to move toward opposite poles of the cell by shortening of microtubules attached to kinetochores.
- 4) **TELOPHASE II:-** The two groups of Haploid chromatids (chromosomes of future cell) once again get enclosed by a nuclear envelope then cytokinesis follows resulting in the formation of tetrad of cells i.e., four haploid daughter cells.

Significance of meiosis

- Meiosis is the process that ensures the conservation of a species-specific chromosome number across generations in sexually reproducing organisms, despite reducing the chromosome number by half.
- It also promotes genetic variability in populations from one generation to the next, which is essential for evolution.

Question [NEET-2023]

Kaun Banayga Nabkaar



Which of the following stages of meiosis involves division of centromere?

1 Anaphase II. ✓

2 Telophase

3 Metaphase I

4 Metaphase II

Question [NEET-2021]

Kaun Banayga Nabkaar



Which of the following stages of meiosis involves division of centromere?

1 Telophase II

2 Metaphase I

3 Metaphase II

4 Anaphase II

Thank
You